

Linux Administration & Security Guide

Comprehensive Documentation for Sysadmins

1. SSH & Remote Access

SSH (Secure Shell) is the industry standard for secure remote management.

Core Commands

- `ssh user@host` : Basic connection.
- `ssh -p 2222 user@host` : Connect using a custom port.
- `scp file.txt user@host:/path/` : Securely copy a file to a remote server.

SSH Key-Based Authentication

Using keys is significantly more secure than passwords. It prevents brute-force attacks.

```
# Generate a modern ED25519 key pair
ssh-keygen -t ed25519 -C "admin@company.com"

# Copy the public key to the server
ssh-copy-id -i ~/.ssh/id_ed25519.pub user@remote-ip
```

SSH Hardening (Config File)

Edit `/etc/ssh/sshd_config` to improve security:

```
PermitRootLogin no
PasswordAuthentication no
```

```
PubkeyAuthentication yes
AllowUsers sysadmin_joe
```

2. Networking Basics

Understanding the flow of data is critical for troubleshooting connectivity issues.

IP Address & Interfaces

Use `ip addr show` to view internal IP addresses and network interface names (e.g., eth0, wlan0).

Testing Connectivity

- **Ping:** `ping -c 5 8.8.8.8` checks if a host is reachable via ICMP.
- **Traceroute:** `traceroute google.com` maps the path packets take through various routers.
- **Netstat/SS:** `ss -tunlp` shows which ports are listening for connections.

3. Firewalls (UFW & Iptables)

UFW (Uncomplicated Firewall)

Designed for ease of use. Ideal for standalone servers.

```
sudo ufw default deny incoming
sudo ufw default allow outgoing
sudo ufw allow 22/tcp
sudo ufw allow 80,443/tcp
sudo ufw enable
```

Iptables Basics

The lower-level tool. Rules are processed in order.

```
# Drop all traffic from a malicious IP
sudo iptables -A INPUT -s 192.168.1.100 -j DROP

# View current rules with line numbers
sudo iptables -L -n -v --line-numbers
```

4. Environment Variables & .bashrc

Persistence via .bashrc

The `~/.bashrc` file is executed for every non-login interactive shell.

```
# Custom Aliases
alias ll='ls -la'
alias logs='cd /var/log && ls -lt'

# Environment Variables
export EDITOR='vim'
export PATH=$PATH:/opt/custom_bin
```

Checking Variables

Use `env` or `printenv` to see all currently exported variables.

5. Log Management

Journalctl (Systemd Logs)

Centralized logging for modern Linux systems.

- `journalctl -xe`: Show most recent logs and explain errors.
- `journalctl -u nginx`: Filter logs for a specific service.
- `journalctl --since "1 hour ago"`: Time-based filtering.

Traditional Logs (/var/log)

- `/var/log/auth.log`: Login attempts and sudo usage.
- `/var/log/syslog`: General system messages.
- `/var/log/apache2/error.log`: Web server errors.

6. Service Management (Systemctl)

The Lifecycle of a Service

```
sudo systemctl start [service]      # Run now
sudo systemctl stop [service]       # Stop now
sudo systemctl restart [service]    # Stop then start
sudo systemctl reload [service]     # Refresh config without downtime
sudo systemctl enable [service]     # Start on boot
sudo systemctl disable [service]    # Do not start on boot
sudo systemctl status [service]     # View current state
```

7. Hardening Linux Security

Best Practices

1. **Update Regularly:** `sudo apt update && sudo apt upgrade -y`.
2. **Fail2Ban:** Protects against brute force. It monitors logs and bans IPs after X failed attempts.
3. **No Root SSH:** Always use a standard user + `sudo`.
4. **Audit Open Ports:** Regularly run `nmap -sV localhost` to see what is exposed.
5. **Secure Shared Memory:** Edit `/etc/fstab` to add `tmpfs /run/shm tmpfs defaults,noexec,nosuid 0 0`.